

EX 11:

Write a program for the task of Credit Score Classification

PROGRAM :

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score
```

```
# Sample dataset
```

```
data = {
```

```
    "Income": [25000, 30000, 45000, 60000, 80000,  
               20000, 75000, 50000, 90000, 35000],
```

```
    "Loan": [10000, 12000, 15000, 20000, 10000,  
             18000, 12000, 25000, 5000, 15000],
```

```
    "Credit_History": [0, 0, 1, 1, 1,  
                       0, 1, 1, 1, 0],
```

```
    "Credit_Score": [  
        "Poor", "Poor", "Good", "Good", "Good",  
        "Poor", "Good", "Average", "Good", "Average"  
    ]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
X = df[["Income", "Loan", "Credit_History"]]
y = df["Credit_Score"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.3,
    random_state=42
)

model = RandomForestClassifier(random_state=42)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual:", list(y_test))
print("Predicted:", list(y_pred))

print("Accuracy:",
      accuracy_score(y_test, y_pred))
```

OUTPUT:

```
Actual: ['Good', 'Poor', 'Poor']
Predicted: ['Good', 'Poor', 'Poor']
Accuracy: 1.0
```